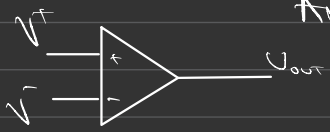
Curs 9



Amplificator Operational

$V_{OUT} = A(V^+ - V^-)$

$A \simeq 10^6 \simeq \infty$

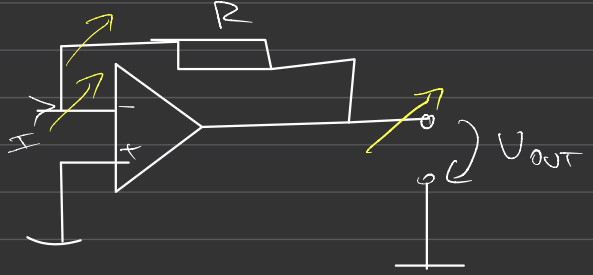
$I^+ \longrightarrow 0$ (rezistență mare, precompensată)

$I^- \longrightarrow 0$

$RN \Rightarrow$ Liniar $(V^+ \simeq V^-)$

- Inversor
- Neinversor
- Sumator
- Logaritmic
- Exponențial

Convertor curent → tensiune



Transimpedanță

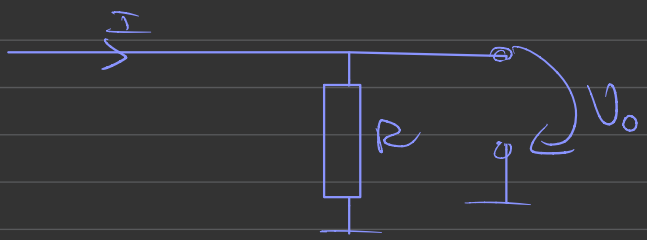
RN $\Rightarrow V^+ = V^-$

$V^+ = 0$

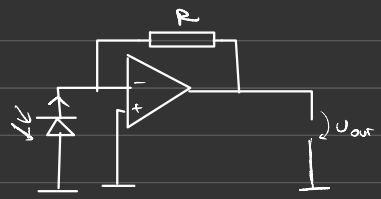
$\Rightarrow V^- = 0$

KCL $\Rightarrow I_N = I_R = \frac{V^- - V_{out}}{R} = \frac{V_{OUT}}{R}$

$V_{out} = -R I_R$

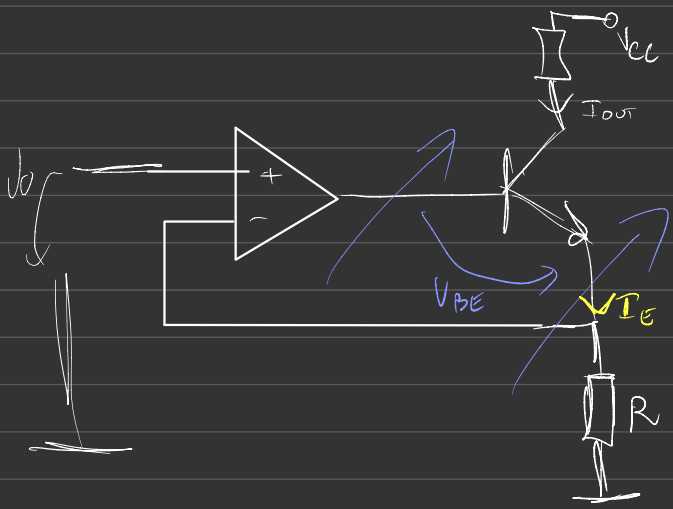


U_{OUT}



Convertor tension → courant

U → I



$V_{BE} = 0,6\,V$

Transistor de tailles

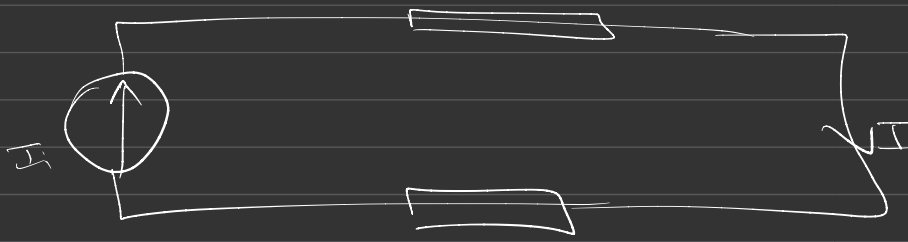
AN ⇒ $V^+ = V^-$

V^+ = tension de consigne (U_i)

$V^- = R \cdot I_E = U_i$

$I_{out} = I_C \simeq I_E = \frac{1}{R} \cdot U_i$



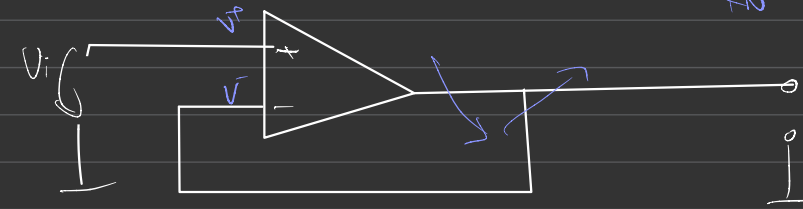


Transmisie în curent

Repetor de tensiune

$U_0 = U_i$

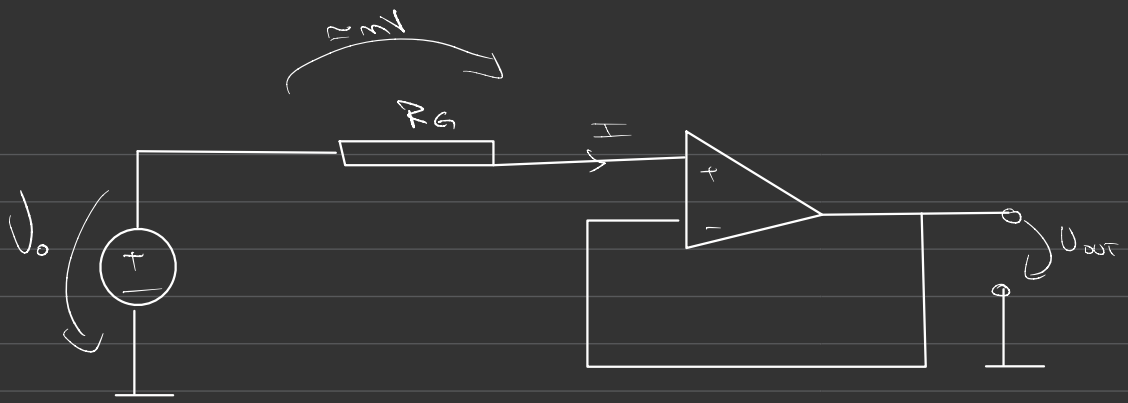
RN $\Rightarrow$ $V^+ = V^-$



$V^- = V^+$

V^- – ieșire

V^+ – tensiune de intrare

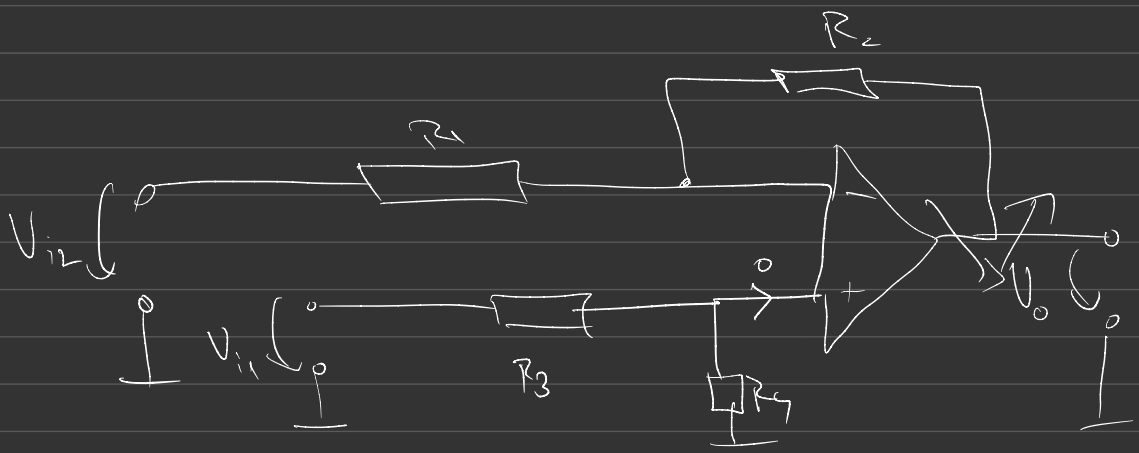


$I = pA$

$R_G = 1k$

Amplificator Diferencial

$$V_o = K(V_{i1} - V_{i2})$$



R.N. $\Rightarrow V^+ = V^-$

$$\frac{U_2 - V^-}{R_1} + \frac{U_0 - V^-}{R_2} = 0$$